What Needs an Authority

Mechanical Detection, Chartered Resolution, and the Exact Price of Sole Ownership

Paper 6 of the Harbor program: the coordination functions of a multi-agent harbor, sorted into those an algorithm discharges and those a charter must pay for

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Abstract

Multi-agent governance documents hand out authorities freely: a *lookout* that scans incoming commitments for conflicts, a *sole owner* for the roadmap, the test suite, the release tag. This paper asks which of those coordination functions genuinely require a designated authority, and prices the ones that do. Two results. First, conflict *detection* needs no authority at all — inside a designed commitment fragment $\mathcal{L}_c$ (definite Horn rules with $\perp$ heads, ground deontic obligation/prohibition rules over scopes and intervals, exclusive interval claims, difference constraints), detection is a polynomial, witness-producing algorithm, $O(H + T \log T + C \log C + V \cdot E)$, validated against a brute-force oracle on 3000 random policy sets with zero disagreements; and composed with the supervisory-control theorem of Paper 2 it is not even a role, but a guard: acceptance is a mediated write, so in-fragment conflict-freedom is *regimentable* — the runtime refuses the commit. One expressive step outside the fragment (disjunctive obligations under discharge choice), conflict-freedom is NP-complete, and *there* an authority becomes necessary: not to notice the conflict but to choose the discharge. Second, sole *ownership* of a role is not a principle but an inequality: a sole specialist beats a pooled swarm exactly when the skill premium clears the Erlang-C boundary $g_A(\rho, c)$ — the whitepaper's proposed threshold is falsified in both directions, with the two thresholds crossing at $\rho = (3 - \sqrt{5})/2$ — and a sole owner who can die and be succeeded is viable at *any* skill premium only while the death-rate–times–succession-time product stays below $D^\star = \eta K/(1 - \eta K)$: past that line, pooling wins even against an infinitely skilled specialist. The slogan the two theorems earn: an authority is needed exactly where the algorithm ends, and where it is needed, its price is a computable number, not a vibe.

Express lane. *Conflict detection inside the designed commitment fragment is polynomial and witness-producing (no judge consumed, and via Paper 2 the check is enforceable pre-commit by the runtime); one disjunctive obligation outside the fragment makes conflict-freedom NP-complete, which is where a chartered resolver must choose; and sole ownership of a role pays iff* $\mu_s/\mu_g \geq g_A(\rho, c)$ *— exact Erlang-C form in the box of §7 — with sole-owner viability capped at* $\xi/\eta \leq D^\star$ *absent skill limits.* Experts: read the two boxes (§4, §7), the succession box (§8), and the inventory table (§9); the rest is scaffolding.

1 The scene: the harbor's authority inventory

A harbor of coding agents runs on commitments: standing prohibitions (*never write to prod*), conditional obligations (*if you deploy, you must update the runbook within the window*), exclusive claims (*this file is mine from tick 5 to tick 20*), deadlines chained to deadlines. The governance document responds with authorities. A *lookout* is chartered to scan every proposed commitment against the accepted set and reject contradictions. *Sole-responsibility roles* — a roadmap owner, a test-suite curator, a release avatar — each hold exclusive write capability over one invariant, on the theory that a single deeply-skilled owner beats a committee.

Both assignments feel natural, and both smuggle in an unexamined premise: that the function *requires a designated authority* — a mind with standing, judgment, and a name — rather than a subroutine and a price. The operations lead's two questions are therefore exact. *Does the lookout need judgment, or is contradiction-scanning mechanical?* And *when does the sole owner actually beat the pool — and what happens when the owner dies mid-quarter?* This paper answers both with theorems that were run before they were believed: the first by a 3000-instance oracle sweep with zero disagreements [internal, `b4_deontic_fragment.py`], the second by a sweep that *falsified the whitepaper's own proposed threshold in both directions* before replacing it [internal, `b8_specialization.py`].

The idea in one breath.

An authority is needed exactly where the algorithm ends: inside a designed commitment fragment, conflict detection is a polynomial witness-producing decision the runtime can enforce pre-commit — no judge; one expressive step outside, conflict-freedom is NP-complete and a chartered resolver must choose; and sole ownership of a role is justified not by principle but by an exact Erlang-C threshold on skill, payable only while the owner's death-rate–times–succession-time stays below $D^\star$.

Intuition: the working port. Watch a real port for a day (Figure 1). Whether two vessels are on a collision course is *geometry* — constant bearing, decreasing range — computed identically by every watch officer; nobody's authority is consulted. Who gives way is next decided by a *rulebook* (the collision regulations) — still mechanical, still nobody's judgment. Only where the rulebook ends — a three-vessel crossing the rules never anticipated — does harbor traffic control exercise chartered discretion. And the port's one genuinely personal authority, the licensed pilot with twenty years on these shoals, is hired against a pool of tug crews only when the water is hard enough to pay the premium — and the port licenses a second pilot, because a port whose only pilot can fall ill has priced its channel's availability at one human body. The structural mapping: collision geometry → in-fragment conflict detection (mechanical, witness-producing); the rulebook's edge → the fragment's NP-complete frontier (where chartered resolution begins); the pilot's premium and the second license → the $g_A(\rho, c)$ boundary and the succession price $D^\star$. The analogy maps relations, so it earns a prediction: making the rulebook *richer* should eventually make rule-following itself intractable and reintroduce discretion — and that is exactly what the NP-completeness theorem says happens one disjunction past the fragment.

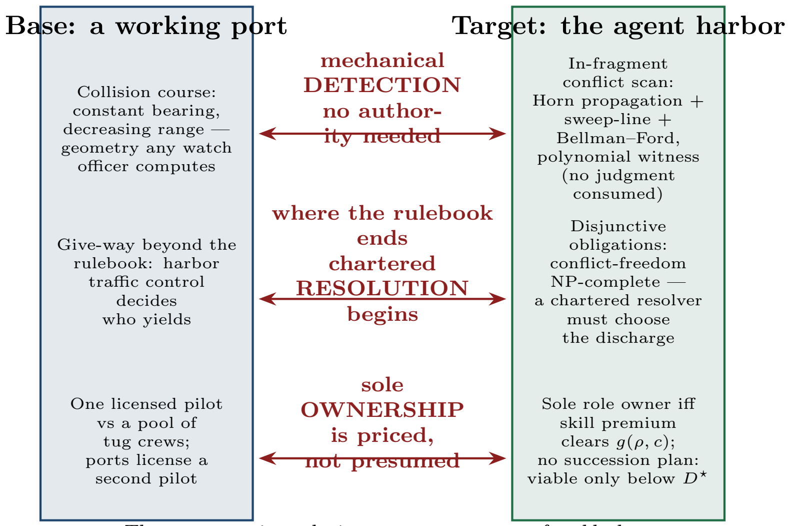
The predictable misread, preempted: *mechanical detection does not mean mechanical governance*. The checker certifies the norm *system*, never the norms — a bad policy consistently held is consistent — and detection without a resolution charter just produces alarms. The theorems do not abolish authority; they *relocate* it, from noticing to choosing, and then hand the remaining authority its bill.

2 The vocabulary, defined

This paper draws on two fields that share no readers. Part I is deontic logic and complexity theory; Part II is queueing theory. A reader fluent in one is very unlikely to be fluent in the other, so both dictionaries are below, in the order the terms are needed. Nothing here is deep; it is all notation someone else fixed decades ago.

Part I: logic and complexity. *Deontic* logic is the logic of obligation and permission, as opposed to the logic of truth — it reasons about what *ought* to be, not what is. Its two operators

Paper 6 — the authority inventory: where the algorithm ends, the charter begins



The map carries relations, not scenery: refusable-by-rule vs decided-by-charter vs paid-for-by-premium.

Figure 1: The relation map. Base domain: a working port — collision-course detection is geometry, give-way beyond the rulebook is harbor traffic control's chartered call, and the sole licensed pilot is paid a premium against a pool of tug crews, with a second license mandated. Target domain: the agent harbor — in-fragment conflict scanning is Horn propagation + sweep-line + Bellman–Ford with a polynomial witness; disjunctive obligations make conflict-freedom NP-complete, so a chartered resolver chooses the discharge; and a sole role owner is justified iff the skill premium clears $g_A(\rho, c)$, with no-succession viability capped at $D^\star$. The correspondence arrows carry the theorem content: refusable-by-rule vs decided-by-charter vs paid-for-by-premium.

here are O (obliged) and F (forbidden), and we apply them only to ground facts: an action, a scope, a time interval. A *Horn rule* is an implication whose premise is a conjunction of atoms and whose conclusion is a single atom — the shape almost every rules engine and every Prolog program already uses. A *definite* Horn program has no negation, which is exactly what makes it cheap: its consequences can be computed by repeatedly firing whatever rules are satisfied until nothing changes, and this process is guaranteed to terminate at one canonical answer. That answer is the *least model*, and its defining property is the one Theorem 1a leans on: it contains precisely the facts forced in *every* consistent world, so a status it assigns is not one possibility among several but a certainty. *Unit propagation* is the name for the firing procedure, and Dowling–Gallier [1] showed it runs in time linear in the program.

A *difference constraint* is an inequality of the form $x_j - x_i \leq c$ over deadline variables — "the runbook update must land no more than six ticks after the deploy." A system of them is satisfiable exactly when its constraint graph contains no *negative cycle*, a loop of constraints whose weights sum below zero and which therefore demands that some event precede itself. Bellman–Ford finds such a cycle if one exists, and hands it over as the witness. This is standard temporal-reasoning machinery; §10 names the literature that owns it.

A problem is in **P** when some algorithm solves it in time polynomial in the input size — the practical meaning of "tractable at scale." It is **NP-complete** when a proposed solution can be *checked* in polynomial time but no polynomial algorithm for *finding* one is known, and finding one would settle the P versus NP question. The working consequence, which is all Theorem 1b needs: an NP-complete check cannot be run on every proposal at commit time as the policy set grows. *3-SAT* is the canonical NP-complete problem (is a Boolean formula in three-literal clauses satisfiable?), and a *reduction* from it is the standard way to prove a new problem at least as hard: encode any 3-SAT instance as an instance of yours, so that solving yours would solve 3-SAT.

A *witness* is the concrete object a checker returns alongside a yes — the derivation chain, the two clashing rules with their overlap, the negative cycle. Producing witnesses is a strictly stronger requirement than deciding, and it is the difference between a checker that says "rejected" and one a human can argue with.

Part II: queueing. An $M/M/1$ *queue* is one server, arrivals landing at random at average rate λ, each job taking a random time averaging $1/\mu$. $M/M/c$ is the same with c interchangeable servers sharing one line — the airport-check-in arrangement, not the supermarket one. *Utilisation* ρ is the fraction of capacity in use; as it approaches 1 waiting time diverges, which is the entire reason these comparisons are not obvious.

The *Erlang-C* formula $C(c, \rho)$ gives the probability that an arriving job finds every server busy and must wait at all. It is a century old [5] and it is the only non-elementary ingredient in Theorem 2; everything else there is algebra on it.

Pooling is the practice of putting all work in one queue served by everyone, as opposed to *dedicating* a specialist to it. The classical fact — and the tension the paper prices — is that pooling wins on waiting time through statistical smoothing, while a specialist may win on service rate through skill. The *skill premium* $r = \mu_s/\mu_g$ is how much faster the specialist is, and Theorem 2 asks exactly how large it must be to pay.

Finally, a queue *with breakdowns* is one whose server fails at some rate and is restored after some delay. Its relevance is that a sole owner is a server that can fail: sessions die, containers are reaped. Theorem 3's W_∞ is the residual delay such a queue carries even with an infinitely fast server, which is why no amount of skill buys it back.

3 Part I. The lookout is a subroutine: the fragment $\mathcal{L}_c$

Cheap detection is a *language-design* choice, not a discovery about deontic logic in general. The commitment language $\mathcal{L}_c$ buys it by construction. A policy set in $\mathcal{L}_c$ contains:

- **facts** (ground atoms) and **definite Horn rules** $a_1 \wedge \cdots \wedge a_k \to b$, where the head b may be the contradiction symbol $\perp$;
- **deontic rules** body $\to \mathsf{M}(\text{act}, s, [t_1, t_2])$ with $\mathsf{M} \in \{\mathsf{O}, \mathsf{F}\}$: when the (Horn) body holds, action act is obliged or forbidden on scope atoms s over the ground time interval $[t_1, t_2]$;
- **exclusive interval claims** (*resource x is exclusively mine on* $[t_1, t_2]$);
- **difference constraints** $x_j - x_i \leq c$ over deadline variables.

A *conflict* is any of four things: derivable $\perp$; a fired O/F pair on the same action with overlapping scope-and-interval; two exclusive claims overlapping on the same resource; or a negative cycle in the difference-constraint graph (an unsatisfiable deadline system). This is deliberately a *ground* fragment — no nesting of deontic operators, no negation-as-failure, monotone bodies only — and §10 says explicitly which classical deontic puzzles are thereby left outside the door.

4 The detection theorem and its frontier

Theorem 1a (membership: detection is polynomial and witness-producing). For a policy set in $\mathcal{L}_c$ with Horn program size H, T fired deontic tokens, C exclusive claims, and difference-constraint graph (V, E), conflict detection is decidable in $O(H + T \log T + C \log C + V \cdot E)$, with a polynomial witness per conflict: Dowling–Gallier unit propagation computes the least Horn model in $O(H)$ — which by monotonicity equals the intersection of all models, i.e. exactly the statuses forced in *every* world [verified framework, Dowling–Gallier 1984]; a keyed sweep-line over interval endpoints finds every O/F clash and claim overlap in $O(T \log T + C \log C)$; Bellman–Ford returns a negative cycle iff the deadline system is infeasible, in $O(V \cdot E)$. The witnesses are, respectively: a derivation chain, the two fired rules with their overlap interval, the two claims, and the negative cycle.

Theorem 1b (frontier: one step outside is NP-complete). Extend $\mathcal{L}_c$ with disjunctive obligations $\mathsf{O}(l_1 \vee \cdots \vee l_k)$ under discharge-choice semantics (the agent picks which disjunct to discharge). Conflict-freedom of a policy set becomes NP-complete: membership by the selection certificate; hardness from 3-SAT — each variable x becomes a pair of deontic rules $\mathrm{sel}_x^+ \to \mathsf{O}(\mathrm{assert}_x)$ and $\mathrm{sel}_x^- \to \mathsf{F}(\mathrm{assert}_x)$ on *identical* scope and interval, each clause becomes a disjunctive obligation over its selector literals, and a conflict-free discharge selection exists iff the formula is satisfiable [verified: reduction restated here in full; checked both directions on 16/16 instances, internal, `b4_deontic_fragment.py`].

Read together, Theorems 1a and 1b are a **dichotomy** at a designed boundary, in the sense Schaefer [10] made standard for satisfiability: on one side of a single expressive choice the problem is polynomial, on the other it is NP-complete, and the language designer is offered nothing in between. We claim the location, not the pattern — the same step is familiar one formalism over, where the simple temporal problem is polynomial and admitting disjunctions of difference constraints makes consistency NP-complete [11]. That is why $\mathcal{L}_c$'s boundary should be read as a price list rather than an accident of this particular fragment.

Verification. 3000 random in-fragment policy sets against a brute-force oracle (all 2^5 Horn models intersected; pairwise overlap checks; exhaustive integer search over the complete potential box $[-\Sigma|c|, 0]^V$ for the difference system): 885 conflicts found — 121 derivable $\perp$, 195 O/F clashes, 484 claim overlaps, 213 negative temporal cycles — of which 149 are reachable *only* through Horn propagation; **0 disagreements** with the oracle, and least model = intersection-of-all-models on all 3000 [internal, `b4_deontic_fragment.py`, seed 20260816]. The 3-SAT reduction was verified in both directions on 8 satisfiable and 8 unsatisfiable random instances (n=4, m=7–28), 16/16, with the satisfying assignment extracted from each conflict-free selection. Mutation: deleting Horn propagation (checking only directly stated statuses) misses 100 of the 885 sweep conflicts — propagation is load-bearing at scale, not a formality [internal].

Numbers by hand. The mutant's canonical miss is a four-line policy a reviewer can check aloud: fact `agent_deployed`; Horn rule `agent_deployed` $\to$ `prod_access`; deontic rule `prod_access` $\to$ $\mathsf{O}(\texttt{write\_prod}, s, [0, 10])$; standing $\mathsf{F}(\texttt{write\_prod}, s, [5, 15])$. Propagation forces `prod_access`, the obligation fires, and the sweep-line reports the O/F overlap on $[0, 10] \cap [5, 15] = [5, 10]$ — while a pairwise scanner that never propagates sees no directly stated clash and stays silent [internal, the printed witness]. For the frontier, run the reduction on the smallest unsatisfiable formula $(x) \wedge (\neg x)$: two disjunctive obligations force both sel_x^+ and sel_x^-, firing $\mathsf{O}(\mathrm{assert}_x)$ and $\mathsf{F}(\mathrm{assert}_x)$ on identical scope and interval — every selection conflicts, matching unsatisfiability [verified: two lines]. *Now you try:* facts {`audit_open`}; rules `audit_open` $\to$ `vault_access` and `vault_access` $\to$ $\mathsf{O}(\texttt{read\_ledger}, s, [2, 6])$; standing $\mathsf{F}(\texttt{read\_ledger}, s, [4, 9])$. Conflict? (Yes: propagation forces `vault_access`, the obligation fires, and the overlap is $[4, 6]$ — a two-hop indirect conflict, exactly the species the mutant misses.)

5 Composition with Paper 2: the lookout is not even a role

Paper 2 proved that a safety policy over an agent's event alphabet is enforceable-by-prevention (*regimentable*) iff it is Ramadge–Wonham controllable — no uncontrollable event carries a legal history out of the policy — and its design rule was: uncontrollable events in a policy's *condition* are free; only uncontrollable events in its *prohibition* are fatal. Now compose. In the harbor, *acceptance of a commitment is a mediated write*: the accepted set lives in the single-writer store, and the acceptance event is a member of Σ_c, the controllable alphabet. The policy *"never accept a proposal whose union with the accepted set conflicts in-fragment"* therefore gates a controllable event on a condition that Theorem 1a makes a total, polynomial-time-computable predicate of committed state. The policy is controllable, hence regimentable: the conflict never enters the accepted set, because the daemon refuses the write, with the theorem's witness attached to the refusal [verified: composition of Theorem 1a with Paper 2's boxed theorem; both components machine-checked separately, internal, `b4_deontic_fragment.py` + `b3_controllability.py`].

This is the paper's first answer, and it is sharper than "the lookout doesn't need judgment." Inside the fragment the lookout is not an authority at all — it is a *guard clause* at the mediation boundary, as impersonal as a type check, producing a four-line witness instead of a verdict. What the charter must still supply is everything the guard cannot: *which* of two conflicting proposals yields (priority, negotiation, waiver), *whether* a standing norm should be amended, and — the moment anyone wants disjunctive obligations — *which discharge to choose*, because Theorem 1b says no proposal-time check scales there unless P=NP. The frontier is thus the exact line where authority re-enters: the schema designer reads Theorem 1 as a price list — keep the language in $\mathcal{L}_c$ and detection is a regimentable subroutine; buy the expressive convenience of disjunction and you have hired a judge, with a queue.

6 Part II. The sole owner: an inequality, not a principle

Part I located the one place inside conflict management where a mind is genuinely required: not to notice a contradiction but to choose a discharge among disjuncts the fragment cannot rank on its own. Part II asks the harbor's other standing authority question, about a different kind of decision entirely — not *which* candidate resolution to pick, but *whether one designated person should be doing the picking at all* — and it gets the same kind of answer: not a principle to invoke but a number to compute. The whitepaper's sole-responsibility roles (roadmap owner, test-suite curator, release avatar) trade pooled capacity for accountability and skill. Model the trade honestly. Requests for the invariant-critical function arrive Poisson at rate λ; a sole specialist serves them as an $M/M/1$ queue at rate μ_s; the pooled alternative is an $M/M/c$ queue of generalists at rate μ_g each, utilization $\rho = \lambda/(c\mu_g)$; the skill premium is $r = \mu_s/\mu_g$. Sole ownership also carries an *accountability value* A (the worth of single-writer attribution per unit time of demand) against a waiting cost w per request-hour. The whitepaper proposed the threshold $\tilde{g} = 1 + (c-1)\rho/(1-\rho)$ and marked it *status: proposed.* The sweep ran before the belief formed — and falsified $\tilde{g}$ in both directions.

7 The specialization boundary

Theorem 2 (queueing specialization boundary). With $C(c,\rho)$ the Erlang-C delay probability, sole ownership weakly dominates the pool in net cost iff the skill premium clears

$$\frac{\mu_s}{\mu_g} \;\geq\; g_A(\rho, c) \;=\; c\rho \;+\; \frac{1}{1 + \dfrac{C(c,\rho)}{c(1-\rho)} + \dfrac{A\mu_g}{w\lambda}},$$

reducing at $A = 0$ to $g(\rho, c) = c\rho + \dfrac{c(1-\rho)}{c(1-\rho) + C(c,\rho)}$, with $g(\rho, 1) = 1$ (a specialist against one generalist needs no premium), $g \to c$ as $\rho \to 1$ (the boundary *saturates* at the capacity ratio, never diverges), and the closed form $g(\rho, 2) = 1 + 2\rho - \rho^2$ [verified: derivation below]. The proposed threshold $\tilde{g} = 1 + (c-1)\rho/(1-\rho)$ is **falsified in both directions**: at

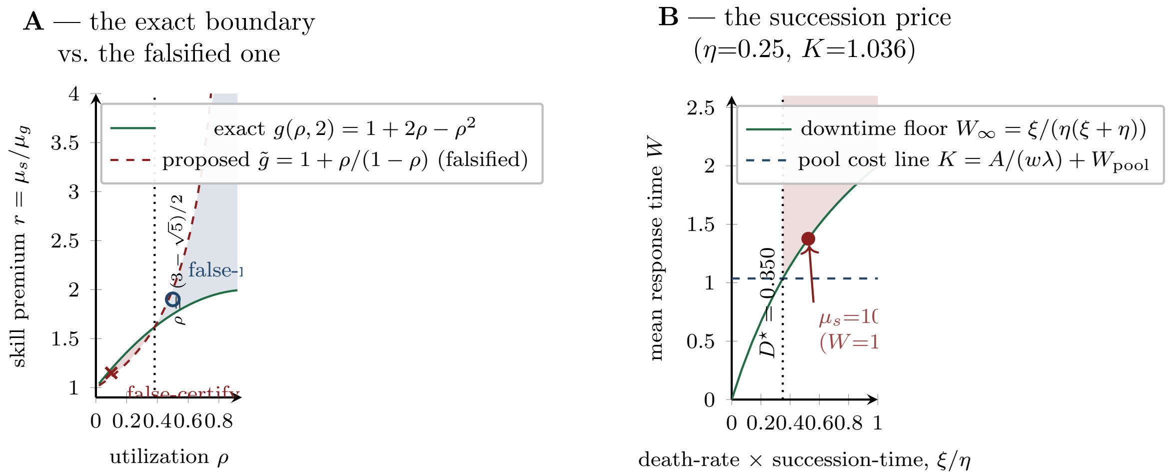


Figure 2: The regime diagram. Panel A: the exact boundary $g(\rho, 2) = 1+2\rho-\rho^2$ (solid) against the falsified proposal $\tilde{g}$ (dashed); the shaded lenses are the two error regimes — false-certify below the crossing at $\rho = (3-\sqrt{5})/2$, false-reject above it — with the two refuting instances marked (r=1.15 at ρ=0.1; r=1.9 at ρ=0.5). Panel B: the succession price at the reference instance (η=0.25, K=1.036): the downtime floor $W_\infty = \xi/(\eta(\xi+\eta))$ crosses the pool cost line at $D^\star = 0.350$; to its right pooling wins at every skill premium — the marked point is a $\mu_s = 10^8$ specialist still losing at $\xi/\eta = 1.5D^\star$ ($W = 1.376 > K$). Both panels regenerate from `b8_specialization.py` [internal, seed 20260816].

c=2, ρ=0.1, r=1.15 it certifies a specialist who loses ($W_{\mathrm{solo}} = 1.0526 > W_{\mathrm{pool}} = 1.0101$; simulation $z = -15.7$), at c=2, ρ=0.5, r=1.9 it rejects a specialist who wins ($1.111 < 1.333$; $z = +32$); the two thresholds cross at $\rho = (3-\sqrt{5})/2 \approx 0.382$, and where g saturates $\tilde{g}$ diverges (ρ=0.9, c=8: $g = 7.73$ vs $\tilde{g} = 64$) [internal, `b8_specialization.py`, seed 20260816].

Numbers by hand. The theorem is two lines of algebra on textbook formulas, which is the right way to first believe it. Sole ownership pays iff $w\lambda(W_{\mathrm{solo}} - W_{\mathrm{pool}}) \le A$, i.e. $1/(\mu_s - \lambda) \le W_{\mathrm{pool}} + A/(w\lambda)$; invert and divide by μ_g, using $\lambda/\mu_g = c\rho$ and $\mu_g W_{\mathrm{pool}} = 1 + C(c,\rho)/(c(1-\rho))$, and g_A falls out [verified]. For $c = 2$ the Erlang-C value is $C(2,\rho) = 2\rho^2/(1+\rho)$, so $c(1-\rho)/(c(1-\rho)+C) = (1-\rho^2)/((1-\rho^2)+\rho^2) = 1-\rho^2$ and $g(\rho, 2) = 2\rho+1-\rho^2$ [verified: three lines]. Now check the falsification instances by hand at $\mu_g = 1$. At $\rho = 0.1$: $g = 1+0.2-0.01 = 1.19$ while $\tilde{g} = 1+0.1/0.9 = 1.111$, so $r = 1.15$ sits between them — the whitepaper certifies, the truth rejects; indeed $W_{\mathrm{solo}} = 1/(1.15-0.2) = 1.0526$ against $W_{\mathrm{pool}} = 1+0.0182/1.8 = 1.0101$ [verified]. At $\rho = 0.5$: $g = 1.75$, $\tilde{g} = 2$, and $r = 1.9$ again sits between — rejected by the whitepaper, but $W_{\mathrm{solo}} = 1/0.9 = 1.111 < W_{\mathrm{pool}} = 4/3$ [verified]. Setting $1+2\rho-\rho^2 = 1+\rho/(1-\rho)$ gives $\rho^2 - 3\rho + 1 = 0$: the thresholds cross at $\rho = (3-\sqrt{5})/2$, which is why the falsification runs in *both* directions — Figure 2, Panel A, shades the two error lenses [verified]. *Now you try:* at c=2, ρ=0.3, does $r = 1.5$ justify a sole owner? ($g(0.3, 2) = 1+0.6-0.09 = 1.51$ — no, by a hair; $\tilde{g} = 1.43$ would have said yes. The flattering threshold flatters exactly below the crossing.)

The cross-validation stack behind the closed forms: Erlang-C means against discrete-event simulation, $|z| \le 1.8$ across the grid; the 60-instance random sweep over $(\lambda, \mu_s, \mu_g, c, A, w, \xi, \eta)$ gave 55 decisive instances and **0** sign violations of the boundary [internal, `b8_specialization.py`].

8 The succession price: what one mortal body costs

The boundary above assumes the specialist never disappears. Sole owners disappear: sessions die, containers are reaped, providers churn. Model the corner honestly: the specialist *dies* at rate ξ and is *succeeded* after an $\mathrm{Exp}(\eta)$ spin-up (context transfer, warm-up, credential rebinding), work resuming where it stopped, requests queueing throughout — an $M/M/1$ queue with breakdowns in the Mitrany–Avi-Itzhak lineage.

Theorem 3 (price of the succession rule). With effective rate $\mu_{\text{eff}} = \mu_s\eta/(\xi+\eta)$, the sole owner's mean response time is exactly

$$W_{\text{bd}} \;=\; \frac{1+\mu_s\xi/(\xi+\eta)^2}{\mu_{\text{eff}}-\lambda} \;=\; \frac{1}{\mu_{\text{eff}}-\lambda} \;+\; \frac{\mu_{\text{eff}}}{\mu_{\text{eff}}-\lambda}\,W_\infty, \qquad W_\infty = \frac{\xi}{\eta(\xi+\eta)},$$

decreasing in μ_s with infimum W_∞: *no amount of skill buys back residual downtime.* Consequently, with $K = A/(w\lambda) + W_{\text{pool}}$ the pool's cost line, pooling dominates at *every* skill premium iff

$$\frac{\xi}{\eta} \;>\; D^\star \;=\; \frac{\eta K}{1-\eta K},$$

and $D^\star$ is the price of the succession rule: a sole role without a succession plan (small η) is viable only while its death-rate–times–succession-time product stays below $D^\star$ [verified: $W_\infty(\xi^\star) = K$ at $\xi^\star = D^\star\eta$ is one line of algebra; closed form checked against matrix-geometric solution to 2×10^{-13}, truncated CTMC to 10^{-14}, Gillespie simulation $|z| \le 0.8$, internal, `b8_specialization.py`].

Numbers by hand. Reference instance: λ=1, c=4, μ_g=1.2 (so $\rho = 0.208$), A=0.2, w=1, η=0.25. Erlang-C gives $C(4, 0.208) = 0.0110$, so $W_{\text{pool}} = 1/1.2+0.0110/(4\cdot1.2\cdot0.792) = 0.8362$ and $K = 0.2 + 0.8362 = 1.0362$ [verified: four-function-calculator arithmetic on the boxed formulas]; then $\eta K = 0.2591$ and $D^\star = 0.2591/0.7409 = 0.350$ [verified]. Just past the line, at $\xi/\eta = 1.5\,D^\star$, an absurdly skilled specialist ($\mu_s = 10^8$) still loses: $W = 1.376 > K = 1.036$ — the floor W_∞, not the skill, is binding [internal]. The identity in the box also explains the program's recorded wrong turn: the naive decomposition $W_{\text{bd}} = 1/(\mu_{\text{eff}}-\lambda)+W_\infty$ is refuted (it predicts 4.17 where the truth is 8.17 at the canary instance μ_s=5, ξ=0.5), because downtime also *builds queue*: the residual is amplified by $\mu_{\text{eff}}/(\mu_{\text{eff}}-\lambda) = 2.5$ there, and $1.5+2.5\times2.667 = 8.17$ checks by hand [verified arithmetic; refutation internal]. Panel B of Figure 2 draws exactly this geometry: the floor W_∞ crossing the pool cost line at $D^\star$. *Now you try:* at the reference instance, is a sole owner with a same-day succession plan ($\eta = 2$; K is unchanged at 1.0362, so $\eta K = 2.07 > 1$) ever capped? (No: $\eta K \ge 1$ makes $D^\star = \infty$ — a fast enough succession plan removes the viability ceiling entirely, leaving only the Theorem-2 premium test. That is what a succession plan *buys*.)

The implementation hazard is also priced. A cost model that forgets breakdowns entirely (plain $M/M/1$) is caught by the high-ξ canary: at $\xi/\eta = 2 \gg D^\star$, the mutant says SPECIALIST while simulation says the pool wins decisively ($\Delta C = -7.06 \pm 0.16$); at low ξ the mutant is indistinguishable — so the canary, not routine testing, is what makes the omission visible [internal, `b8_specialization.py`].

The predictable misread, preempted: $D^\star$ does not say specialists are bad, and Theorem 2 does not say pooling is the default truth. Together they say sole ownership is a *purchase*: below the boundary the premium does not cover the congestion cost and the honest ledger entry is "we pay latency $w\lambda(W_{\text{solo}}-W_{\text{pool}})-A$ per unit time for accountability"; above $D^\star$ the purchase is unavailable at any price until the succession plan (larger η) is bought first. Every quantity in both tests — λ, μ_s, μ_g, ξ, η — is daemon-metered, so the harbor can re-grade its roles from telemetry rather than from taste.

9 The authority inventory, priced

The two parts assemble into the answer to the title question:

Coordination function	Authority needed?	What it costs
In-fragment conflict detection	none (algorithm)	$O(H + T\log T + C\log C + V\cdot E)$ per batch, witness included
In-fragment conflict enforcement	none (runtime guard)	mediated acceptance write; regimentable via Paper 2
Conflict-freedom with disjunctive obligations	chartered resolver	NP-complete check, or a judge who chooses the discharge
Priority, waiver, norm amendment	chartered, irreducibly	outside every checker; the charter's residual job
Sole ownership of a role	conditional	justified iff $r \geq g_A(\rho, c)$; otherwise a priced latency purchase
Sole ownership without succession	conditional, capped	viable only while $\xi/\eta \leq D^\star = \eta K/(1-\eta K)$

The first two rows delete an authority the governance document thought it needed; the third and fourth rows locate the authority that is genuinely irreducible; the last two rows keep the authority but hand it an invoice. That is the paper's claim in institutional terms: a well-designed harbor spends judgment *only* where theorems certify judgment is required, and meters what the remaining judgment costs.

10 Related work

Imported. Linear-time Horn satisfiability and unit propagation are Dowling–Gallier [1]; we import the algorithm and the least-model property unchanged, and our fragment is *designed* so that it applies. NP-completeness machinery is Cook and Karp [2, 3]; the reduction in Theorem 1b is standard 3-SAT gadgetry, and the contribution is only the *location* of the frontier — discharge-choice disjunction — in a commitment language a scheduler would actually want. The regimentation composition imports Ramadge–Wonham supervisory control [4] via Paper 2. On the queueing side, the Erlang-C delay formula is a century old [5], the economies of pooling in many-server queues are classical folklore sharpened by Halfin–Whitt [6], and queues with server breakdowns descend from Mitrany–Avi-Itzhak [7]; Theorem 3's W_{bd} is in that lineage. **Positioned against deontic logic.** The fragment is ground and operator-nesting-free: conflicts are defined operationally (O vs F on overlapping scope and interval), not as theoremhood in standard deontic logic [8], and the classical paradoxes — Ross's paradox, Chisholm's contrary-to-duty puzzle [9] — simply cannot be *stated* in $\mathcal{L}_c$: that is a designed retreat, declared rather than hidden, and it is what the tractability is paid for with. **New, honestly.** No component algorithm or queueing formula above is ours. The contributions are: the fragment/frontier pair as an exact price list for commitment-language design; the composition making in-fragment conflict-freedom regimentable rather than merely checkable; the falsification and replacement of the whitepaper's specialization threshold by the exact $g_A(\rho, c)$ with its $c{=}2$ closed form and crossing point; and the succession criterion $D^\star$ stating, in daemon-metered quantities, when no skill premium can rescue a sole role. An August-2026 survey found no prior statement of the authority question in this detection/resolution/ownership decomposition; a final lit-sweep before submission is owed — "not found" is not "proven nonexistent."

11 Honest boundary

What these theorems do NOT say. (i) Theorem 1's completeness is relative to least-model semantics: monotone bodies only — negation-as-failure, unification, and symbolic interval endpoints coupled to difference variables all leave the fragment, and nothing here is claimed about them. (ii) The batch complexity bound is what is proven; the incremental amortized form (watched literals, interval trees) is a constants engineering claim, stated in the portfolio, not certified here. (iii) The checker certifies the norm *system*, never the norms: a harbor whose policies are consistently harmful passes every scan — norm content is chartered authority's irreducible job, row four of the inventory. (iv) The regimentation composition inherits Paper 2's deployment obligation wholesale: acceptance must *actually* be completely mediated (Σ_c honest), which a red-team, not this proof, discharges. (v) Theorem 2 compares *mean* costs only — no tail percentiles, no SLA quantiles — under FCFS and exponential everything; A and w are policy prices the theorem constrains but does not pick. (vi) Theorem 3's breakdown model is death-at-all-times with preemptive resume (the death/heartbeat model); active-only breakdowns would let sufficient skill escape the W_∞ floor, so the model choice is load-bearing and declared. (vii) The wrong turns are on record and priced: the naive W_{bd} decomposition is refuted (4.17 vs 8.17), and the whitepaper's $\tilde{g}$ was falsified by this program's own sweep — the correction, not the original, is what ships. (viii) The inventory covers detection, resolution, and ownership; other authority functions (charter amendment, dispute appeal, norm authorship) are chartered by construction and no theorem here bears on them.

Reproducibility. Every [internal] number regenerates from two self-contained scripts at seed 20260816: `skills/harbor-results/scripts/b4_deontic_fragment.py` (3000-instance oracle sweep, conflict census, least-model check, 16/16 reduction verification, Horn-propagation mutation) and `skills/harbor-results/scripts/b8_specialization.py` (falsification instances, Erlang-C vs DES, matrix-geometric / CTMC / Gillespie cross-validation of W_{bd}, 60-instance sign sweep, $D^\star$ instance, breakdown-forgetting mutation canary). Every [verified] number is a closed form or short derivation restated in this paper's boxes and checkable by hand or from the cited literature. The two figures regenerate from `docs/harbor-research/figures/src/paper6_figures.py`.

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